

KAI ZHEN

<http://kaizhen.us> • kaizhen723@gmail.com

1. EMPLOYMENT

Full-Time

Amazon.com, Inc.

2021 – Now

- Senior Applied Scientist @ Amazon, Inc.
 - AGI / Foundation Models, Multimodal + Inference Efficiency, Seattle, WA
 - Operated dense and MoE LLM training across hundreds of GPUs; optimized parallelism, sharding strategies, optimizer dynamics, and failure recovery.
 - Owned post-training efficiency work (quantization, pruning, compression) with direct impact on inference latency and serving cost.
 - Drove software-hardware co-design efforts for edge/cloud Alexa deployment.
 - Published and patented work adopted in production systems.
- Applied Scientist II @ Alexa AI
 - Alexa Hybrid Science, Pittsburgh, PA
 - Led the release of multiple on-device ASR models, making accuracy/latency tradeoffs for production deployment.

[Blog Post Highlight on Amazon Science Website \(LLM Compression Track\)](#)

Internship

Amazon.com, Inc.

- Applied Scientist Intern
 - Alexa Speech, Pittsburgh, PA
 - Project: Network Compression for On-Device Speech Recognition

Summer 2020

[Best Internship Poster Presentation Award](#)

LinkedIn Corporation

- Machine Learning & Relevance Intern
 - Ads-AI Group, Mountain View, CA
 - Company Standardization Group, New York City, NY

Summer 2019

Summer 2018

Academic Part-Time

Indiana University

Aug. 2015 – Mar. 2021

- Research Assistant: Audio Signal Analysis/Synthesis Technology Based on Machine Learning
 - Published in leading machine learning and speech processing conferences and journals
 - Contributed to 5 US patents as an inventor
- Associate instructor in Department of Computer Science and Intelligent Systems Engineering

2. EDUCATION

Ph.D., in Computer Sciences and Cognitive Science

May. 2021

- Indiana University, Bloomington, United States
- Committee: Minje Kim (chair, IU Intelligent Systems Engineering), Robert Goldstone (co-chair, IU Cognitive Science), Donald Williamson (IU Computer Science), and Shen Yi (U. of Washington, Speech and Hearing Sciences)
- Dissertation: "[Neural Waveform Coding: Scalability, Efficiency and Psychoacoustic Calibration](#)"
[Winner of the Outstanding Research Award \(IU Cognitive Science\)](#)

M.S., in Computer Science

Jul. 2015

- Tsinghua University, Beijing, China

B.S., in Software Engineering

Jul. 2012

- Xidian University, Xi'an, China
-

3. PROFESSIONAL ACTIVITIES

Conference Reviewer

- IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP): 2019 - 2025
- ISCA Interspeech: 2022 - 2025

- Annual Meeting of the Association for Computational Linguistics (ACL): 2025
- Conference on Empirical Methods in Natural Language Processing (EMNLP): 2025
- ICLR Workshop on Sparsity in LLMs (SLLM): 2025
- EURASIP European Signal Processing Conference (EUSIPCO): 2022 - 2023
- IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA): 2021 - 2023
- IEEE International Conference on Data Mining (ICDM): 2020
- Association for the Advancement of Artificial Intelligence (AAAI): 2017 - 2018

Journal Reviewer

- European Association for Signal Processing (EURASIP) Journal on Audio, Speech, and Music Processing
- IEEE MultiMedia
- Speech Communication

4. PUBLICATIONS

Referred International Conference Proceedings

- [C015] Jiajun Zhou, Yifan Yang, **Kai Zhen**, Ziyue Liu, Yequan Zhao, Ershad Banijamali, Athanasios Mouchtaris, Ngai Wong, Zheng Zhang, "QuZO: Quantized Zeroth-Order Fine-Tuning for Large Language Models," in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP 2025), Main Conference, Suzhou, China, November 4-9, 2025*.
- [C014] Zhen Zhang, Yifan Yang, **Kai Zhen**, Nathan Susanj, Athanasios Mouchtaris, Siegfried Kunzmann, Zheng Zhang, "MaZO: Masked Zeroth-Order Optimization for Multi-Task Fine-Tuning of Large Language Models," in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP 2025), Main Conference, Suzhou, China, November 4-9, 2025*.
- [C013] Ryan Solgi, **Kai Zhen**, Rupak Vignesh Swaminathan, Nathan Susanj, Athanasios Mouchtaris, Siegfried Kunzmann, Zheng Zhang, "Saten: Sparse Augmented Tensor Networks for Post-Training Compression of Large Language Models," in *Findings of the Empirical Methods in Natural Language Processing (EMNLP) 2025, Suzhou, China, November 4-9, 2025*.
- [C012] Yifan Yang*, **Kai Zhen***, Bhavana Ganesh, Aram Galstyan, Goeric Huybrechts, Markus Müller, Jonas M. Kübler, Rupak Vignesh Swaminathan, Athanasios Mouchtaris, Sravan Babu Bodapati, Nathan Susanj, Zheng Zhang, Jack FitzGerald, Abhishek Kumar, "Wanda++: Pruning Large Language Models via Regional Gradients," in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL), Vienna, Austria, July 27-August 1, 2025*.
* Co-first authors.
- [C011] Yifan Yang*, **Kai Zhen***, Bhavana Ganesh, Aram Galstyan, Goeric Huybrechts, et al., "Fast and Effective N:M LLM Pruning via Block-wise Calibration," in *Proceedings of the ICLR Workshop on Sparsity in Large Language Models (ICLR-SLLM), Singapore, 2025*.
* Co-first authors.
- [C010] Yifan Yang, **Kai Zhen**, Ershad Banijamal, Athanasios Mouchtaris, Zheng Zhang, "AdaZeta: Adaptive Zeroth-Order Tensor-Train Adaption for Memory-Efficient Large Language Models Fine-Tuning," in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP), Miami, USA, 12-16 November, 2024*.
- [C009] Rupak Vignesh Swaminathan, Grant Strimel, Ariya Rastrow, Harish Mallidi, **Kai Zhen**, Hieu Nguyen, Nathan Susanj, Athanasios Mouchtaris, "Max-Margin Transducer Loss: Improving Sequence-Discriminative Training Using a Large-Margin Learning Strategy," in *Proceedings of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), Seoul, Korea, 14-19 April, 2024*.
- [C008] Martin Radfar, Paulina Lyskawa, Brandon Trujillo, Yi Xie, **Kai Zhen**, Jahn Heymann, Denis Filimonov, Grant Strimel, Nathan Susanj, Athanasios Mouchtaris, "Conmer: Streaming Conformer with no self-attention for interactive voice assistants," in *Proceedings of the Annual Conference of the International Speech Communication Association (Interspeech), Dublin, Ireland, August 21-24, 2023*.
- [C007] **Kai Zhen**, Martin Radfar, Hieu Duy Nguyen, Nathan Susanj, Grant Strimel, Athanasios Mouchtaris, "Sub-8-bit Quantization for On-Device Speech Recognition: A Regularization-Free Approach," *IEEE Workshop on Spoken Language Technology (IEEE SLT), Doha, Qatar, January 9-12, 2023*.
- [C006] **Kai Zhen**, Hieu Duy Nguyen, Raviteja Chinta, Nathan Susanj, Athanasios Mouchtaris, Tariq Afzal, and Ariya Rastrow, "Sub-8-Bit Quantization Aware Training for 8-Bit Neural Network Accelerator with On-Device Speech Recognition," in *Proceedings of Annual Conference of the International Speech Communication Association (Interspeech), Incheon, Korea, September 18-22, 2022*.
- [C005] **Kai Zhen**, Hieu Duy Nguyen, Feng-Ju (Claire) Chang, Athanasios Mouchtaris, "Sparsification via Compressed Sensing for Automatic Speech Recognition," in *Proceedings of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), Toronto, ON, Canada, June 6-12, 2021*.

- [C004] Haici Yang, **Kai Zhen**, Seungkwon Beack, Minje Kim, "Source-Aware Neural Speech Coding for Noisy Speech Compression," in *Proceedings of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, Toronto, ON, Canada, June 6-12, 2021.
- [C003] **Kai Zhen**, Mi Suk Lee, Jongmo Sung, Seungkwon Beack, and Minje Kim, "Efficient and Scalable Neural Residual Waveform Coding with Collaborative Quantization," in *Proceedings of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, Barcelona, Spain, May 4-8, 2020.
- [C002] **Kai Zhen**, Mi Suk Lee, Minje Kim. "A Dual-Staged Context Aggregation Method towards Efficient End-To-End Speech Enhancement," in *Proceedings of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, Barcelona, Spain, May 4-8, 2020.
- [C001] **Kai Zhen**, Jongmo Sung, Mi Suk Lee, Seungkwon Beack, and Minje Kim, "Cascaded Cross-Module Residual Learning towards Lightweight End-to-End Speech Coding," in *Proceedings of Annual Conference of the International Speech Communication Association (Interspeech)*, Graz, Austria, September 15-19, 2019.

International Journal Articles

- [J002] **Kai Zhen**, Jongmo Sung, Mi Suk Lee, Seungkwon Beack, and Minje Kim, "Scalable and Efficient Neural Speech Coding: A Hybrid Design," *IEEE/ACM Transactions on Audio, Speech, and Language Processing (IEEE/ACM TASLP)*, **30** (2021): 12-25.
- [J001] **Kai Zhen**, Mi Suk Lee, Jongmo Sung, Seungkwon Beack, and Minje Kim, "Psychoacoustic Calibration of Loss Functions for Efficient End-to-End Neural Audio Coding," *IEEE Signal Processing Letters (SPL)* **27** (2020): 2159-2163.

Patents

- [P005] Kim, Minje, Mi Suk Lee, Seung Kwon Beack, Jongmo Sung, Tae Jin Lee, Jin Soo Choi, and **Kai Zhen**. "Apparatus and method for speech processing using a densely connected hybrid neural network." U.S. Patent Application 17/308,800, filed November 11, 2021.
- [P004] Mi Suk Lee, Seung Kwon Beack, Jongmo Sung, Tae Jin Lee, Jin Soo Choi, Minje Kim, **Kai Zhen**. "Method and apparatus for processing audio signal," *US Patent App. 17/156,006*, 2021.
- [P003] Minje Kim, **Kai Zhen**, Mi Suk Lee, Seung Kwon Beack, Jongmo Sung, Tae Jin Lee, Jin Soo Choi. "Residual coding method of linear prediction coding coefficient based on collaborative quantization, and computing device for performing the method." U.S. Patent Application 17/098,090, filed May 13, 2021.
- [P002] Mi Suk Lee, Jongmo Sung, Minje Kim, **Kai Zhen**. "Audio signal encoding method and audio signal decoding method, and encoder and decoder performing the same," U.S. Patent Application No. 16/543,095
- [P001] Minje Kim, Aswin Sivaraman, **Kai Zhen**, Jongmo Sung, et al., "Audio signal encoding method and apparatus and audio signal decoding method and apparatus using psychoacoustic-based weighted error function", *US Patent Application*, US 2019 / 0164052 A1.

Peer Reviewed Workshops & Forums

- [W005] **Kai Zhen**, Martin Radfar, Hieu Duy Nguyen, Nathan Susanj, Grant Strimel, Athanasios Mouchtaris. General Quantization for On-Device ASR. *Amazon Machine Learning Conference (AMLC)*, 2022.
- [W004] **Kai Zhen**, Hieu Duy Nguyen, Feng-Ju (Claire) Chang, Athanasios Mouchtaris. Network Sparsification for On-Device ASR. *Amazon Machine Learning Conference (AMLC) Workshop on Network Inference Optimization*, 2020.
- [W003] **Kai Zhen**, Aswin Sivaraman, Jongmo Sung, Minje Kim. On Psychoacoustically Weighted Cost Functions Towards Resource-efficient Deep Neural Networks for Speech Denoising. *The 7th Annual Midwest Cognitive Science Conference*, 2018.
- [W002] Peter Miksza, Kevin Watson, **Kai Zhen**, Sanna Wager, Minje Kim. Relationships between experts' subjective ratings of jazz improvisations and computational measures of melodic entropy. *The Improvising Brain III: Cultural Variation and Analytical Techniques Symposium*, Atlanta, GA, in Feb, 2017.
- [W001] **Kai Zhen** and David Crandall. Finding egocentric image topics through convolutional neural network based representations (extended abstract). In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR) Workshop on Egocentric Computer Vision*, 2016.

5. HONORS, AWARDS & SCHOLARSHIP

- | | |
|---|----------------------|
| Outstanding Research Award | Apr. 2021 |
| <ul style="list-style-type: none"> Given by <u>Cognitive Science Program at Indiana University</u> | |
| Top-Rated Intern Poster | Aug. 2020 |
| <ul style="list-style-type: none"> <u>Among 17 interns receiving the highest rate out of more than 180 participants</u> | |
| Summa Cum Laude | Jul. 2012 |
| <ul style="list-style-type: none"> Graduate with honor from Xidian University | |
| China National Scholarship | Nov. 2010, Nov. 2011 |
| <ul style="list-style-type: none"> For the effort on maintaining top-tier GPA and mathematical contest in modeling (MCM) | |

6. INVITED TALKS

- [Microsoft Research Talks](#), September, 2020 [[Video link](#)]
 - Indiana University Hearing Sciences Seminar, March, 2019
 - Indiana University Grey Matters, Graduate and Post-doc Colloquium, March, 2019
-